Soham Nighojkar

408-892-4258 | sohamnigh@gmail.com | linkedin.com/in/soham-nighojkar | github.com/SohamNigh | sohamnigh.github.io

Experience

Philips — Cambridge, MA
Software Engineering Intern — *January 2026 – June 2026*

- Achieved 100% audit trace coverage across 10+ patient monitoring microservices by redesigning NATS JetStream's advisory subscription model with message subject filtering to catch previously missing logs.
- Eliminated manual metric registration across 20+ microservices by building a self-registering Prometheus system in C# (thread-safe singleton, idempotent deduplication), letting 75+ NuGet packages register metrics independently.
- Cut deployment time 16% and removed manual version bumping by consolidating 18+ fragmented Git workflows across 15 repos into one CI/CD pipeline with release-please versioning and automated Docker tagging.
- Delivered real-time change detection on patient health data from 1,000+ connected devices by deploying Helm-managed Kubernetes clusters onto Philips HealthSuite's NATS messaging infrastructure.

Observe.ai — Redwood Shores, CA
Software Engineering Intern — *June 2025 – August 2025*

- Reduced client-side timeouts 58% by capping per-client request rates on a 10K request/second internal Spring Boot API with a token-bucket limiter (local cache, Redis/Redisson fallback), so excess requests failed fast instead of queueing past their timeout.
- Cut stale WebSocket connections 22% across 1,000+ concurrent sessions by rejecting excess connections during the handshake instead of after accepting them, tracking live counts with an atomic Redis counter shared across replicas.
- Delivered sub-3-second transcription latency across 5,000+ calls by streaming live WebSocket audio through a Kafka pipeline to Deepgram STT, fanning transcripts out to multiple downstream consumers off a single stream.

Trillo Inc. — San Ramon, CA
Software Engineering Intern — *June 2023 – August 2023*

- Accelerated document processing 84% by building Vertex AI pipelines on Google Cloud that parsed PDFs with PyMuPDF and ran named entity recognition to extract structured fields at 92% accuracy.
- Held read latency under 150ms and reduced backend load 27% by building a document store on Google Cloud Blob Storage with indexed queries and batched write operations.

Projects

Mini Inference Serving System | *Go, C++, gRPC, Protocol Buffers, Prometheus* — *August 2026 – Present*

- Served model inference over gRPC from a Go request layer fronting a C++ compute engine, cutting per-request backend calls 75% under concurrent load via dynamic batching (goroutines/channels, size-8 or 10ms window) with context deadlines preserving per-request timeouts.
- Achieved 3ms median CPU latency per batch of 8 by hand-writing matmul, ReLU, and softmax kernels in C++ with no ML framework, loading weights from a custom binary export of a 784-128-10 network trained in NumPy.

Crypto Anomaly Detector | *Python, Kafka, PySpark, PyTorch, Airflow, FastAPI* — *June 2026 – August 2026*

- Flagged Binance wash trading in real time by streaming from a multi-threaded producer through 5 Kafka (KRaft) topics into PySpark Structured Streaming, deriving four rolling features served over FastAPI.
- Tuned the alert threshold with a precision-recall sweep maximizing F1 under a 5% false-positive cap on a 3-layer PyTorch MLP with class-weighted BCE, retrained by an Airflow DAG that promotes only non-regressing models (within 0.01 F1) and flags feature drift weekly.

Education

Northeastern University — Expected: May 2028
Candidate for B.S. in Computer Science — GPA: 3.8/4.0
Honors: Khoury College Dean's List
Relevant Coursework: Computer Systems, Algorithms & Data Structures, Object-Oriented Design, Theory of Computation, Logic and Computation, Discrete Structures
Activities: TAMID Group Tech Consulting, NU Impact, Northeastern Club Basketball

Technical Skills

Languages: Python, Go, Java, C#, C++, TypeScript, JavaScript, SQL
Infrastructure & Data: AWS, GCP, Docker, Kubernetes (Helm), gRPC/Protobuf, Kafka, NATS, Redis, PostgreSQL, Airflow, Spark, Prometheus, Grafana
Frameworks & Libraries: FastAPI, Spring Boot, .NET, Node.js, PyTorch, Pandas, NumPy
Systems: Distributed Systems, Event-Driven Architecture, Streaming Pipelines, Concurrency, Observability, CI/CD